

CAPE SUCKLING BUOY 46082, AK, USA

WMO: 997204

Lat: 59.681N

Lon: 143.372W

Elev: 0

StdP: 101.33

Time Zone: -10.00 (W10)

Period: 05-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	-4.1	-2.9	N/A	N/A	N/A	N/A	N/A	N/A	21.4	1.9	19.2	3.0	7.3	0	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	1.5	15.5	N/A	15.1	N/A	14.7	N/A	N/A	N/A	N/A	N/A	N/A	N/A	4.1	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.8	17.2	15.6	-5.1	16.2	1.5	1.1	-6.2	17.0	-7.0	17.7	-7.9	18.3	-9.0	19.1		
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.1	2.9	2.9	2.5	4.6	7.4	10.5	12.7	13.1	11.3	8.1	4.8	4.1
	DBStd	4.29	2.82	2.45	2.32	1.54	1.81	1.73	1.48	0.96	1.36	1.70	2.29	2.00
	HDD10.0	1313	221	199	231	162	82	13	1	0	3	61	154	184
	HDD18.3	4097	479	432	489	412	338	234	174	162	210	318	404	443
	CDD10.0	258	0	0	0	0	3	29	85	97	43	1	0	0
	CDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH23.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0
Wind	WSAvg	6.8	8.1	7.3	7.6	6.5	5.5	5.1	4.7	5.3	7.0	7.2	8.2	8.5
Precipitation	PrecAvg	1623	158	132	103	87	84	75	110	151	196	191	165	172
	PrecMax	1969	289	200	211	156	151	146	154	233	319	293	302	266
	PrecMin	1102	47	53	45	43	28	29	52	51	99	72	70	79
	PrecStd	185	55	42	43	29	35	23	25	54	56	61	61	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.0	7.9	7.0	10.0	12.9	15.4	16.3	16.0	14.7	11.8	9.7	8.4
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	DB	7.4	6.9	6.5	8.3	11.7	14.5	15.6	15.4	14.2	11.1	9.0	7.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	DB	6.7	6.6	6.0	7.2	10.7	13.7	15.2	15.0	13.8	10.6	8.5	7.1
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	DB	6.1	6.0	5.5	6.5	10.1	13.0	14.8	14.6	13.3	10.2	7.8	6.6
		MCWB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	2%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	10%	WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCDB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Mean Daily Temperature Range	5% DB	MDBR	2.5	2.3	2.4	1.9	1.6	1.4	1.4	1.5	1.6	2.0	2.7	2.3
		MCDBR	1.9	2.4	2.0	2.3	2.2	1.9	1.8	2.0	1.7	1.8	2.5	2.1
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	5% WB	MCDBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		MCWBR	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Clear-Sky Solar Irradiance	taub	0.289	0.317	0.331	0.376	0.381	0.381	0.376	0.378	0.354	0.335	0.299	N/A	
	taud	2.199	2.333	2.369	2.256	2.323	2.387	2.450	2.442	2.469	2.345	2.168	N/A	
	Ebn at Noon	491	693	804	821	845	852	848	817	775	654	455	N/A	
	Edh at Noon	43	65	85	113	113	108	100	94	77	64	44	N/A	
All-Sky Solar Radiation	RadAvg	0.28	0.92	2.24	3.94	5.01	5.16	4.48	3.45	2.36	1.14	0.41	0.17	
	RadStd	0.04	0.14	0.26	0.32	0.38	0.33	0.39	0.47	0.30	0.13	0.05	0.02	

Historical Trends

	DBAvg	Heating		Cooling		Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page